LU Decomposition

$E \cdot A = U$

$A = LU$

E^{-1}

$$\begin{bmatrix} 2 & 1 \\ 8 & 4 \end{bmatrix}$$ → singular, not invertible.

$$\begin{bmatrix} 2 & 1 \\ 8 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix}$$

A → U

$$E = \begin{bmatrix} 1 & 0 \\ -4 & 1 \end{bmatrix}$$

$EA = U$

$A = E^{-1}U$

$= LU$

$$E^{-1} = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}$$

Lower triangular

$A = LU$

$$\begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}$$

always 1

L

$$\begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix}$$

to get 1

U

$2+0 \quad 1+0$

$0 \quad -1$

$$\begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{bmatrix}$$

L D U

D: Diagonal

if u mutiply the diagonals → get the determinant of original matrix.

A → caveats of LU decomp → no row swapping

3×3

1st step → 0 in 21 pos

2nd → 0 " 31 pos

3rd → 0 " 32 "

$$E_{32}E_{31}E_{21}A = (E)^{-1}U$$

3×3

consider identity for now

$$= \left(E_{32}E_{31}E_{21}\right)^{-1} = E_{21}^{-1}E_{31}^{-1}E_{32}^{-1}$$

LIFO

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -5 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

So, E_{21}^{-1} E_{32}^{-1}

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 5 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 5 & 1 \end{bmatrix}$$

L

only works with square matrix.

* can singular matrix decompose into LU?

no inverse

↓

0 in pivot yet no row swap allowed

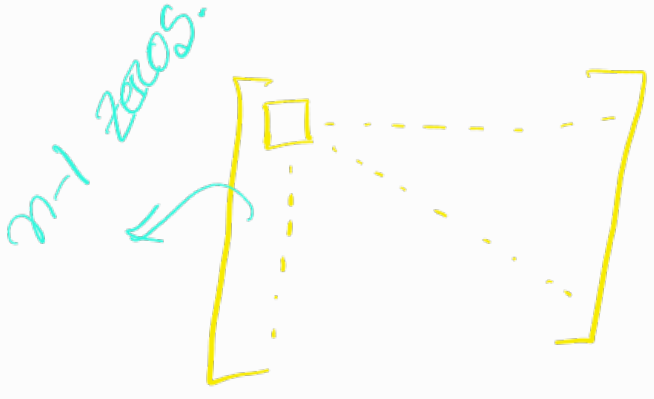
⇒ [☑ ☑ ☐] → can't do a row swap.

Computational complexity of this:

of operation

ops = (mult + add)

$\approx n^2$



$\approx n^2 + (n-1)^2 + \cdots 2^2 + 1^2$

(multiplying & adding → n^r times.

↳ $\frac{n(n+1)(2n+1)}{6}$ roughly n^3 operations for $n \times n$ MAT.

What happens when we need to do a rowswap

↓

=> $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ → take permutation.

3×3 P_{13}

↳ 6 matrix like this → 3!

for $n \times n$ → $n!$ different permutation.

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

P_{13} P_{13} P_{13}^{-1}

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} = I$$

P^{-1} P

$P^{-1} = P^T$ → Transpose.

$(ABC)^T = C^T B^T A^T$ [H.W; Why?]

Symmetric Matrices:

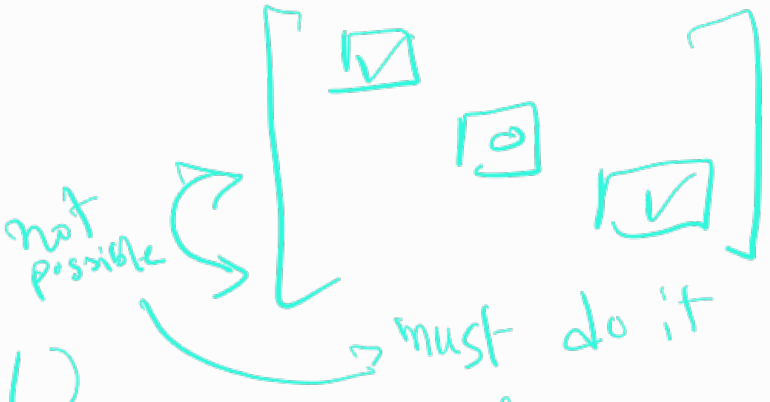
$$A = \begin{bmatrix} 3 & 7 & -3 \\ 7 & -2 & 5 \\ -3 & 5 & 1 \end{bmatrix} \rightarrow A^T = A$$

A^TA → Always be symmetric.

$$(A^TA)^T = A^TA = A^T(A^T)^T = A^TA$$

rowswap formula:

=>



Normally $A = LU$

$\therefore P \cdot A = LU$

↓ permutation MAT